

Frequently Asked Questions for Intro. CFD (4RM00)

Which code can I use for the final assignment?

You can either use the code we supply or your self-written code. Any commercial software or open-source code is not allowed.

What's the difference of index i, j and index I, J ?

Due to staggered grid, we use I, J to assign the scalar variables such as P, T , and i, j to assign the vector variables u, v . It is actually not necessary, as you can see in the code that they always have the same values ($i=I, j=J$). However we purposely differentiate them for you to better and easier understanding of staggered grid.

How shall I choose the grid size or time step?

The flow solvers are only stable up to Courant–Friedrichs–Lewy (CFL) number of 1. This means $CFL = \frac{u_{max}\Delta t}{\Delta x} < 1$. You can use your inlet velocity as an estimation for u_{max} . In practice, you should set CFL number much smaller than 1 in your simulation.

Why my simulation results do not get better or closer to exact values when I refine my mesh?

When you refine your mesh (increasing NPI, NPJ), check if your results converge or not? Usually you also need to increase the iteration number to ensure the convergence.

My simulation give 'NAN' output. What can I do?

There can be many reasons. But most probably it's due to divergence of the solver. So you can try: use smaller timestep Δt and/or lower the relaxation factors. If this doesn't help, you need to check carefully the implementation of your code or boundary conditions.

I can't find all the boundary conditions in the bound.m file.

Most of the boundary conditions are straightforwardly implemented in the bound.m function file. However, some boundary conditions, particularly for internal walls are more convenient to be implemented elsewhere, e.g. in ucoeff.m, vcoeff.m, Tcoeff.m. Some boundary conditions (i.e. fixed value boundary conditions) are automatically satisfied through the initialization, because values in the walls are not updated.

My simulation takes very long time. How can I make it faster?

Our experience tells that mostly your "long time" is not really long; it's normal that your simulation can take up to an hour. The computational time depends largely on the total number of grid cells ($NPI*NPJ$). So you can either make your computational domain smaller or enlarge the grid size.

When do I need to use a turbulence model?

You need to make an estimation of the Reynolds number for your flow problem. For pipe flows, turbulent flow means $Re > 2000$. For other cases, the values are different (search online).

Note that the $k-\epsilon$ model has been implemented in the code. However, this model is not always the best choice depending on your real problem.

When I plot a 2D image of my results, the domain is flipped or rotated or I get an error message “Matrix dimensions must agree.”

This requires an understanding on how matrix and plotting functions work in Matlab. In Matlab a matrix $T(m,n)$ is defined by m rows and n columns. Note that it is first rows and then columns. This is different from the notation usually used in CFD, where $T(I,J)$ refers to the temperature in respectively x and y direction, i.e. first columns and then rows. This means that the order of the rows and the columns is swapped with respect to the standard definition in Matlab. What to do?

To swap the rows and columns back, you can simply transpose the matrix, i.e. $T2 = T'$, where $T2(n,m)$ is the transpose of $T(m,n)$.

So, if you want to plot your results, you can use the following commands:

```
surf(x,y,T')      %surface plot of the temperature
quiver(x,y,u',v') %vector plot of the velocity field
pcolor(x,y,T')    %contour plot of the temperature
```

Additionally, a proper combination of functions `view(...)`, `flipud(...)`, `transpose(...)` usually also works.